

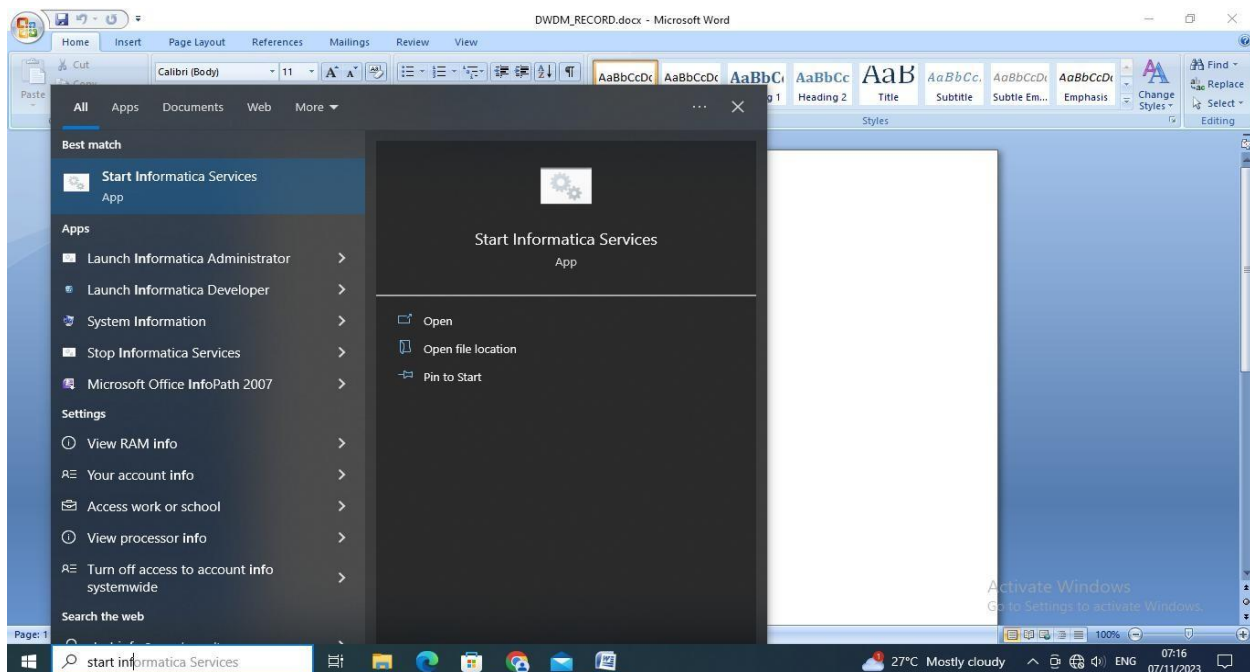
PREREQUISITES OF DATA WAREHOUSING USING INFORMATICA POWER CENTER EXPRESS (PERSONAL EDITION) TOOL

I. Installing the Tool:

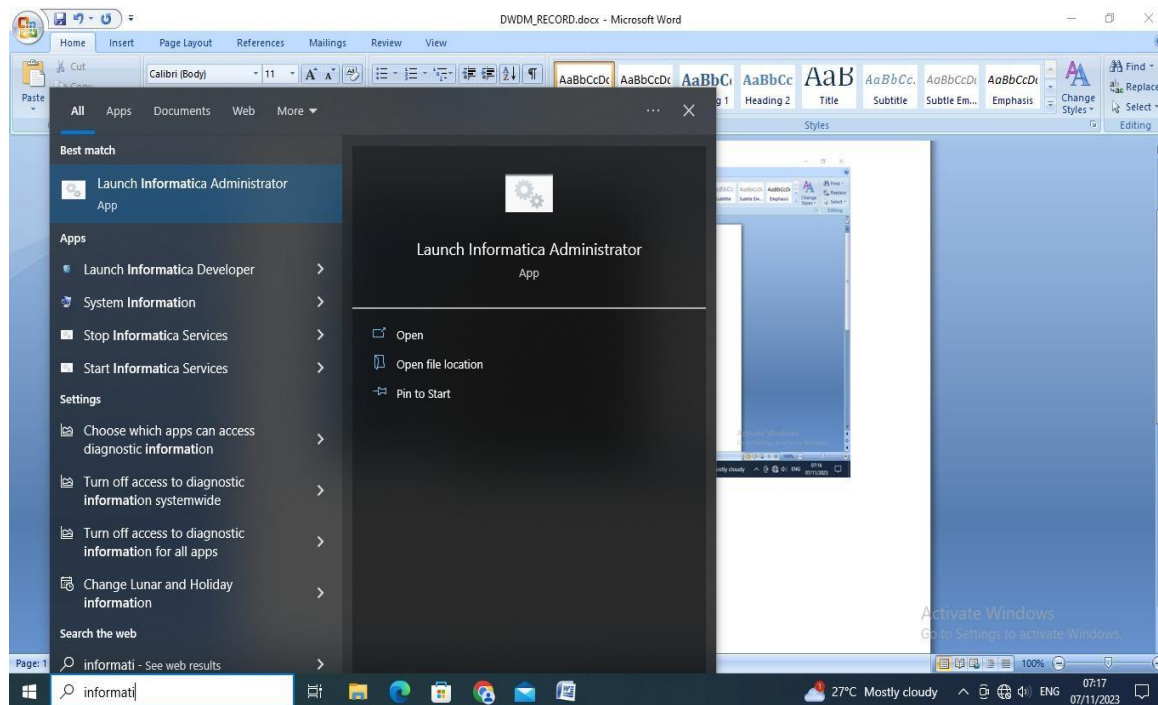
Make sure that you have the latest version of Java installed on your system and the PATH variable set in the System Environment Variables sub-section of Advanced System Settings of your computer. Then install Informatica Power Center Express (Personal Edition) version 9.6.1 as per the instructions given by the manual. Remember the username and password that you entered for the server while installation as these settings will be required for the tool to work properly.

II. Starting Informatica Power Center Express

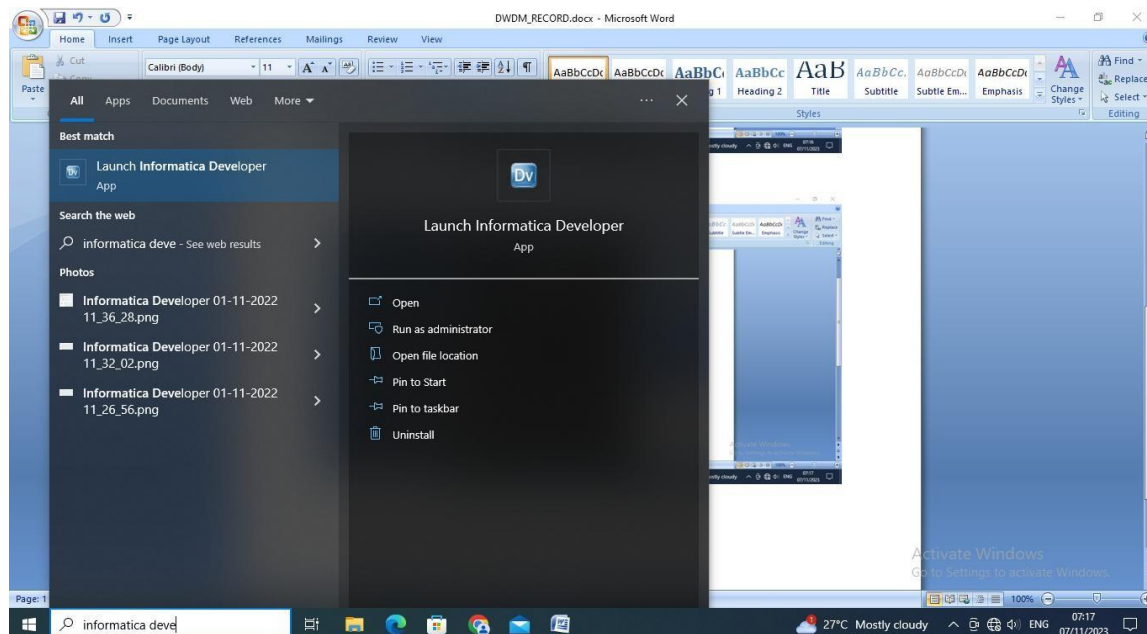
Step 1: After booting the system up, go to Start Menu and run **Start Informatica Services** with Administrator privileges.



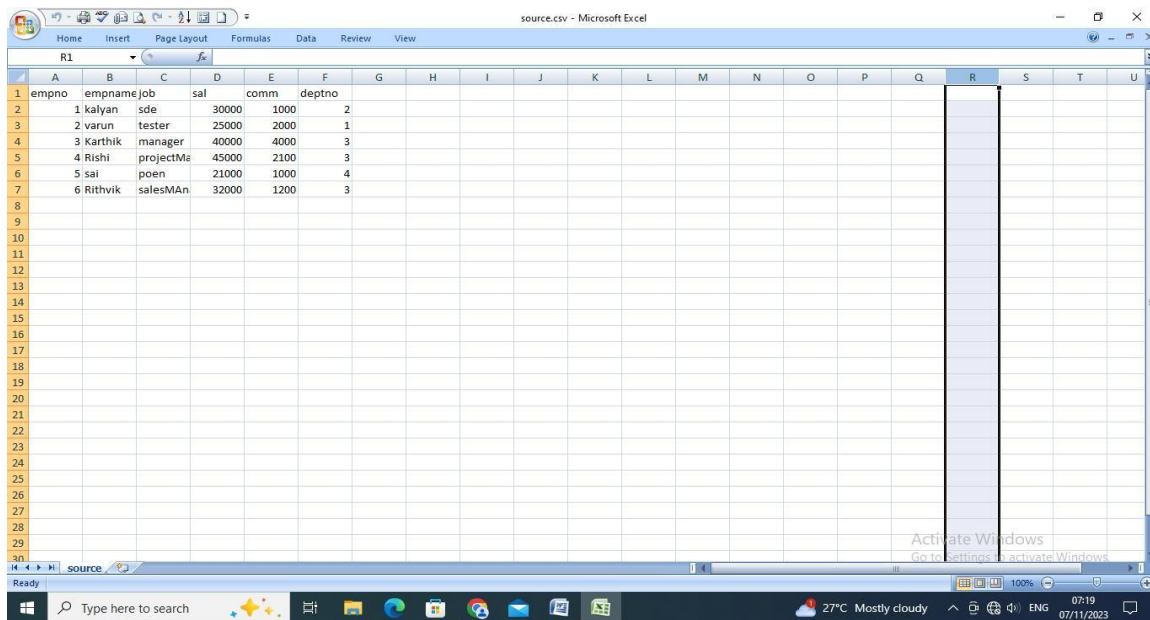
Step 2: Once the batch file runs its course, navigate to **localhost:7009** in your favorite browser. This is the login page for the Administrator Server. Login with the credentials that were supplied during the install process.



Step 3: Once the server page starts up, we are ready to **Launch Informatica Developer** from the Search Bar.

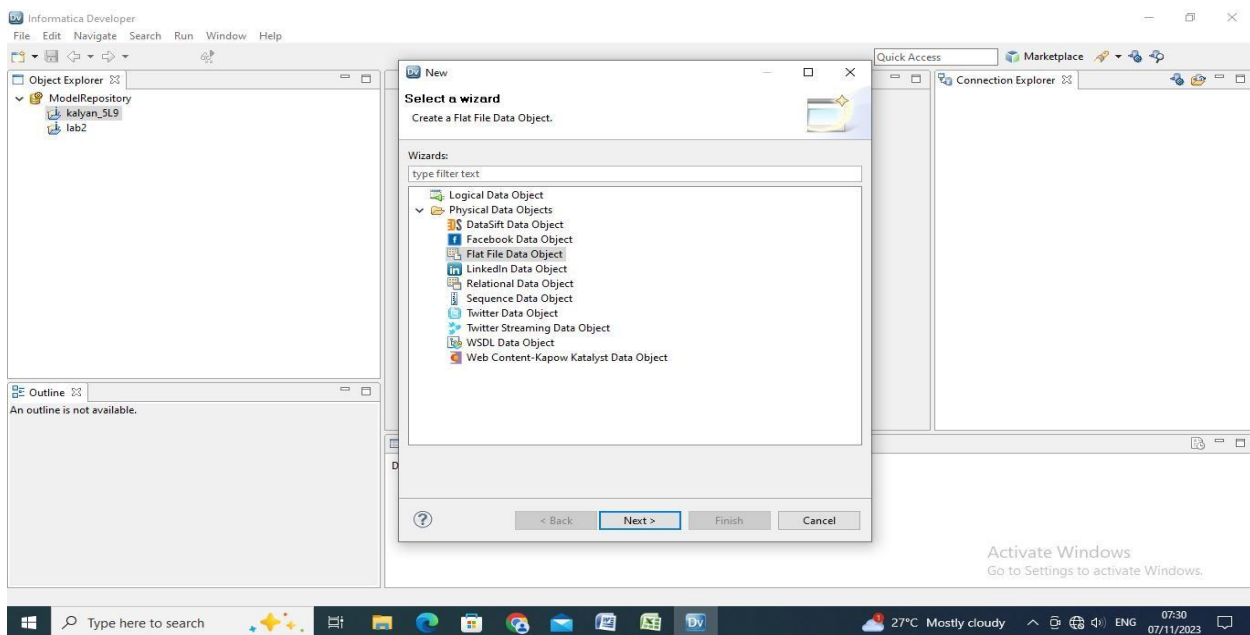


SOURCE TABLE:

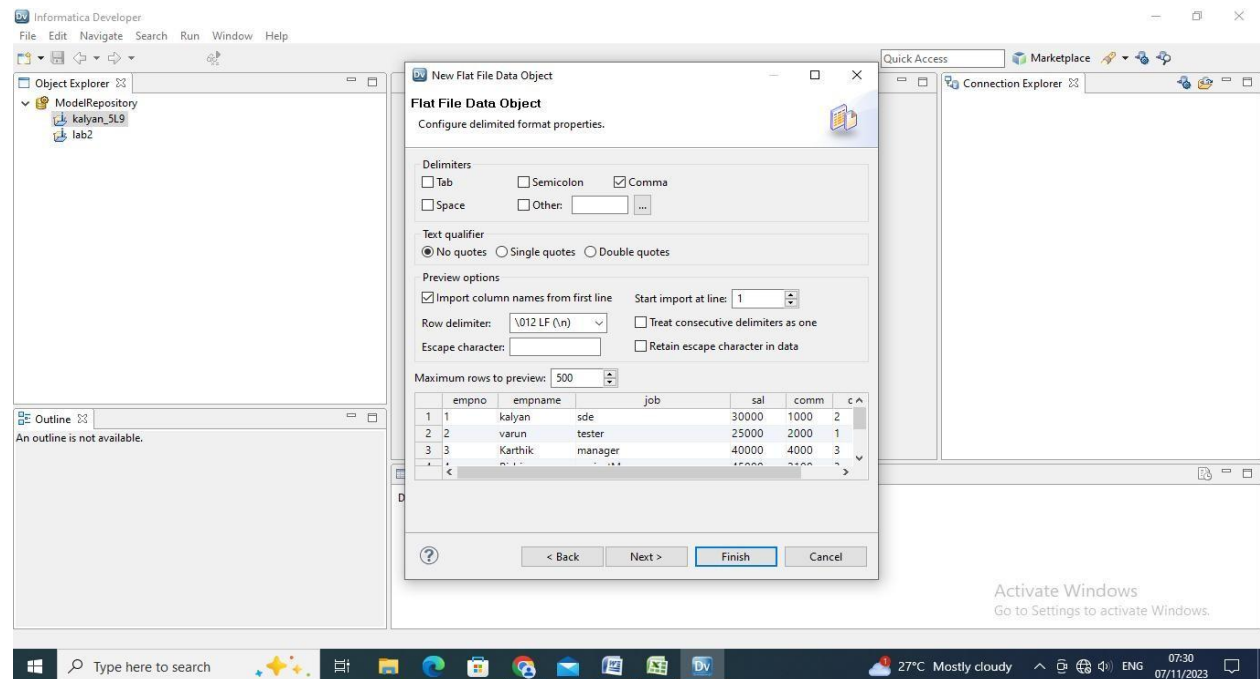
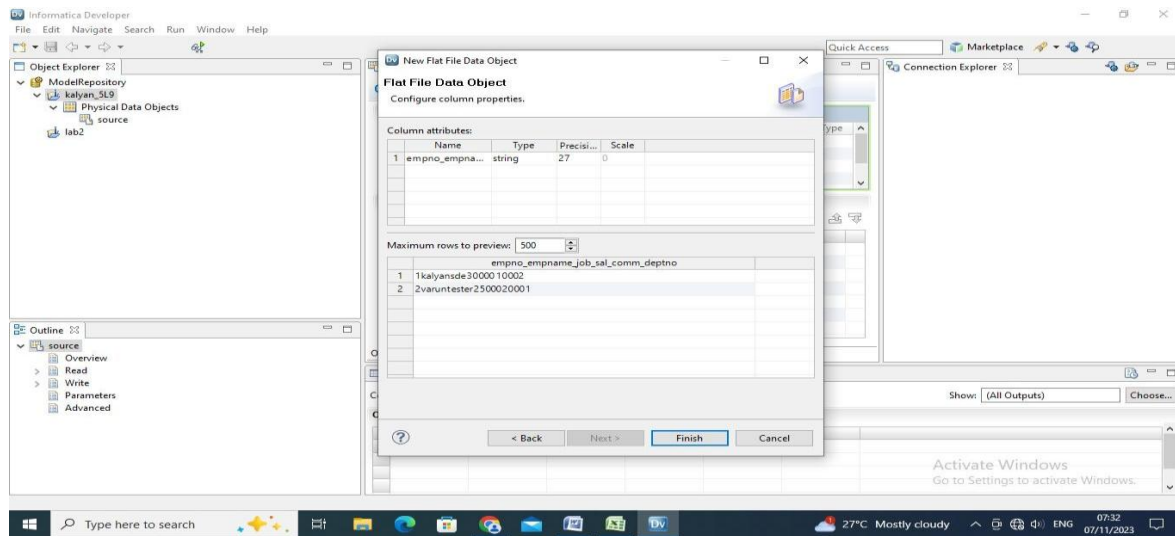


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	empno	empname	job	sal	comm	deptno															
2	1	kalyan	sde	30000	1000	2															
3	2	varun	tester	25000	2000	1															
4	3	Karthik	manager	40000	4000	3															
5	4	Rishi	projectMa	45000	2100	3															
6	5	sai	poen	21000	1000	4															
7	6	Rithvik	salesMAN	32000	1200	3															
8																					
9																					
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Step 4: Once Informatica Developer loads up, connect to Model Repository using the credentials as specified in the server and you are good to go.



IMPORTING SOURCE FILE:



IMPORTING TARGET FILE:

1. Perform filter transformation for the employee database.

Aim: To apply Filter Transformation to a Data Set and filter out tuples matching a certain criterion.

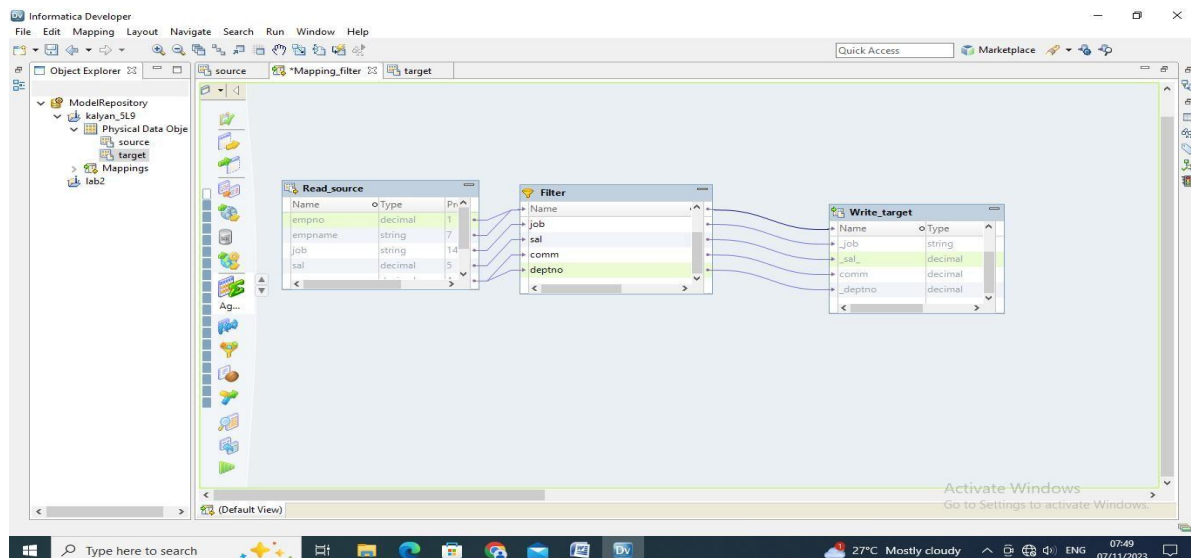
Theory: The filter transformation takes one table as an input and writes into another table all the values that match a certain criterion. The tuples which do not meet the criteria are left as they are.

Procedure:

Step 1: Create the source and target flat files (text files), with the first row being column name and subsequent rows being records, separated by ',' as a delimiter.

Step 2: Start up Informatica Services and Launch Informatica Developer.

Step 3: Import your flat files (both source and target files) into **Vaishnavi_N1** project. Click on Next and then Next again. In the next dialog box that opens, check the **"Import Column Names from the first line"** checkbox and make sure the delimiter selected is ','. Then click on **Finish**.



Step 4: Once the files have been imported, create a new mapping and name it **Filter_Mapping**.

Step 5: Import the source file with read mode and target file as write mode and drag and drop the Filter Transformation from the Palette into the workspace. Give the necessary connections.

Step 6: Make sure that **Filter** is selected. Then navigate to **Properties** → **Filter** and specify the condition in the **Filter Condition** box. Click on **Validate Expression** at the right corner of the Properties Tab, to check the validity of the Filter Condition.

Step 7: Click on the **Data Viewer** Tab of the Properties Explorer and select Run. If the Transformation is successful, then all entries matching the condition are displayed. Then click on **Run** in the menu bar and select **Run Mapping**.

Step 8: Make sure that the connections are intact. Now Run the Mapping. Then navigate to **"F:\Informatica\PCExpress\tomcat\bin\target"** (that's where Informatica is installed on my system, F: Drive), and you can find your output target file. Open it up to see the output.

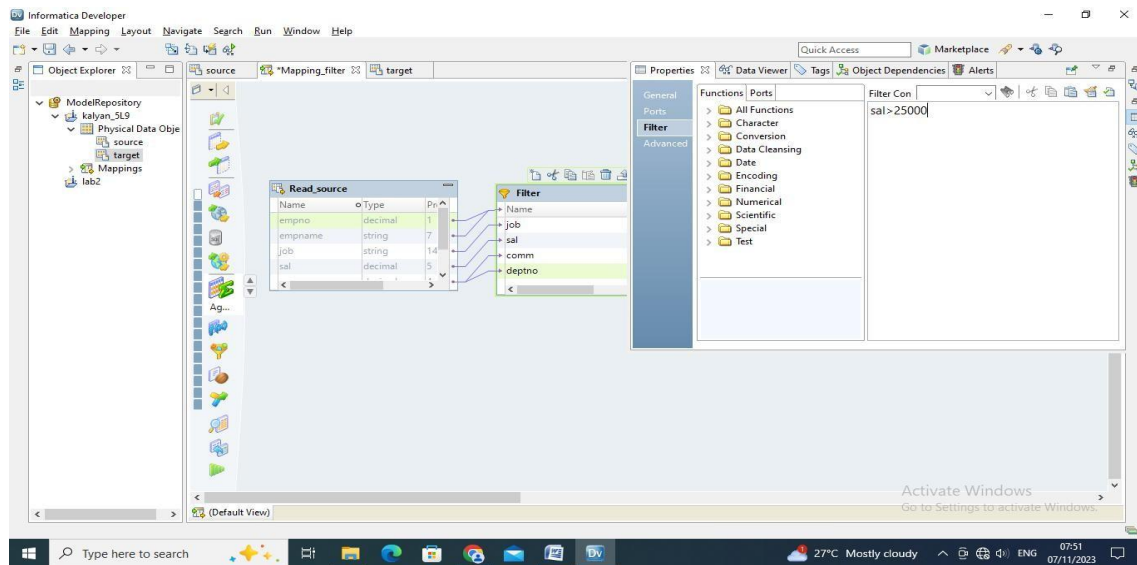
OUTPUT:

The screenshot shows the Informatica Developer interface. On the left, the Object Explorer displays the 'ModelRepository' with a 'Physical Data Object' named 'kalyan_5L9'. The main workspace shows a mapping transformation with a 'Read_source' component connected to a 'Filter' component. The 'Filter' component is configured with a condition on the 'sal' field. The 'Data Viewer' tab on the right displays the output of the transformation, showing a table with 6 rows of data. The table has columns: empno, empname, job, sal, comm, and deptno. The data is as follows:

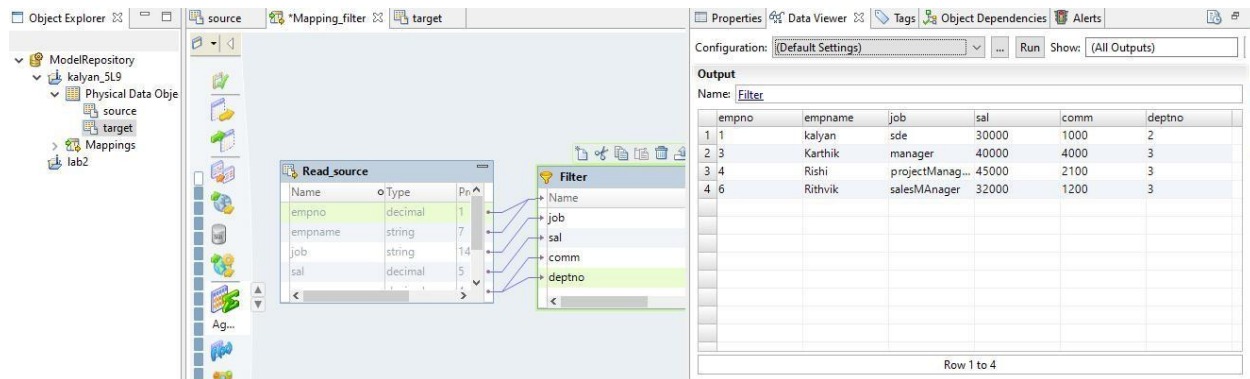
empno	empname	job	sal	comm	deptno
1	kalyan	sde	30000	1000	2
2	Varun	tester	25000	2000	1
3	Karthik	manager	40000	4000	3
4	Rishi	projectManag...	45000	2100	3
5	sai	poem	21000	1000	4
6	Rithvik	salesManager	32000	1200	3

The status bar at the bottom indicates 'Row 1 to 6' and 'Activate Windows Go to Settings to activate Windows.' The system tray shows the date and time as 07:48 on 07/11/2023.

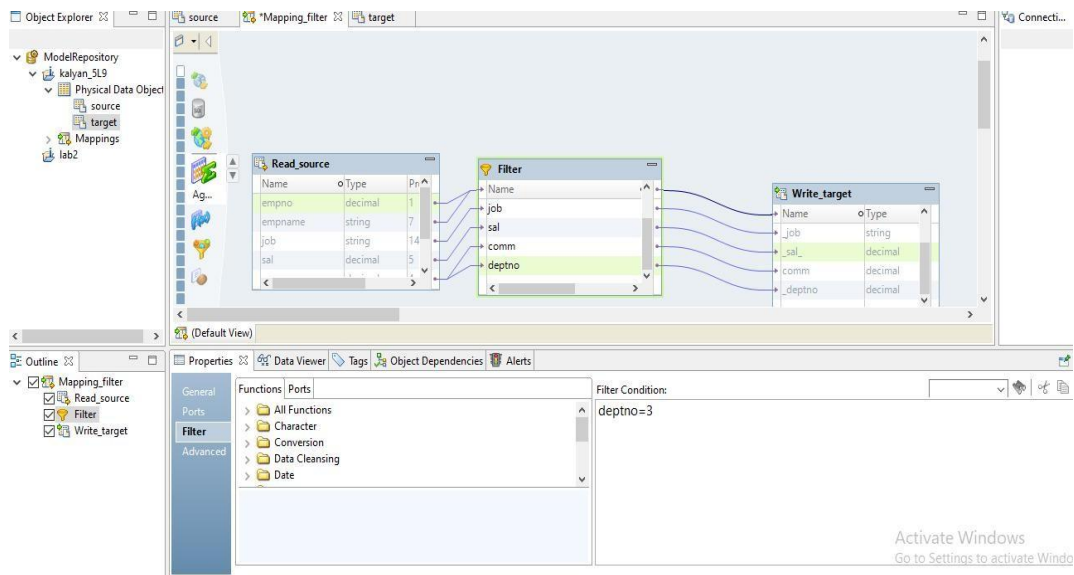
SAL>25000 (INPUT):



OUTPUT: 1) DEPTNO=3



(INPUT):



OUTPUT:

ModelRepository

kalyan_SL9

Physical Data Obj...

source

target

Mappings

lab2

Ag...

Read_source

Name	Type	Pr
empno	decimal	1
empname	string	7
job	string	14
sal	decimal	5

Filter

Name
job
sal
comm
deptno

Write_target

Name	Type
job	string
sal	decimal
comm	decimal
deptno	decimal

(Default View)

Properties

Data Viewer

Tags

Object Dependencies

Alerts

Configuration: (Default Settings)

Run

Show: (All Outputs)

Choose

Output

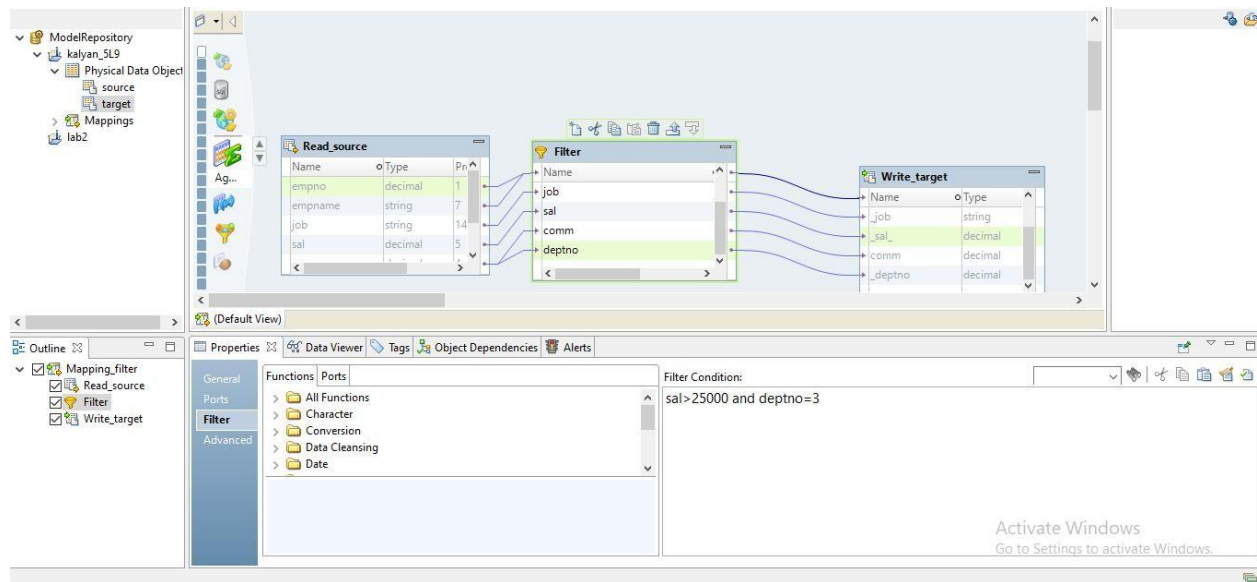
Name: Filter

empno	empname	job	sal	comm	deptno	
1	3	Karthik	manager	40000	4000	3
2	4	Rishi	projectManag...	45000	2100	3
3	6	Rithvik	salesManager	32000	1200	3

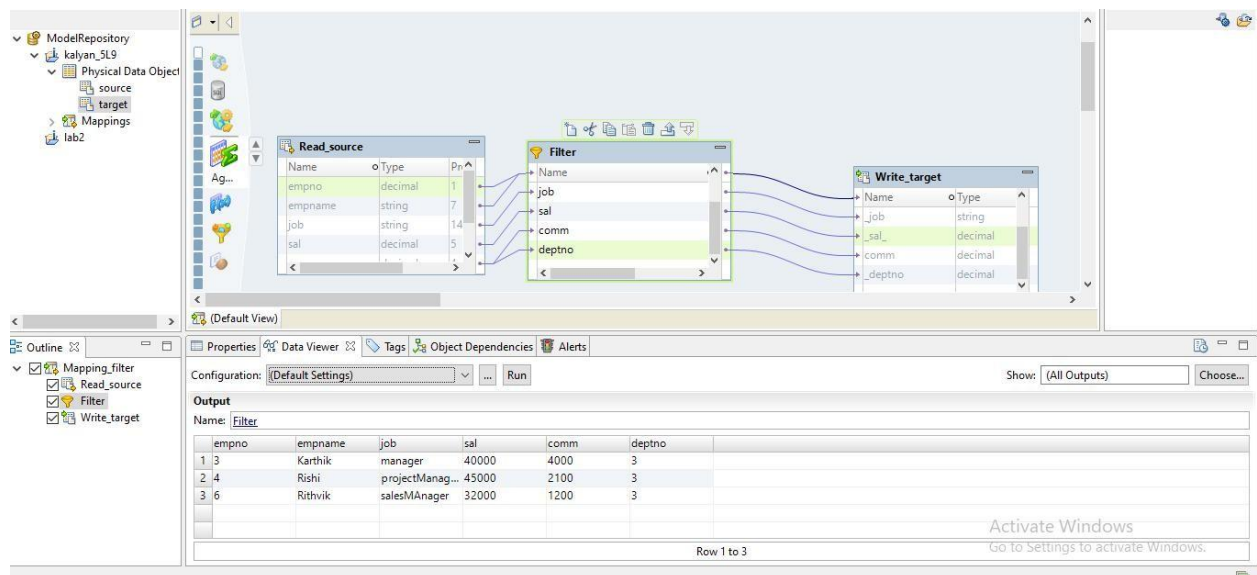
Activate Windows

2) sal>25000 and deptno=3

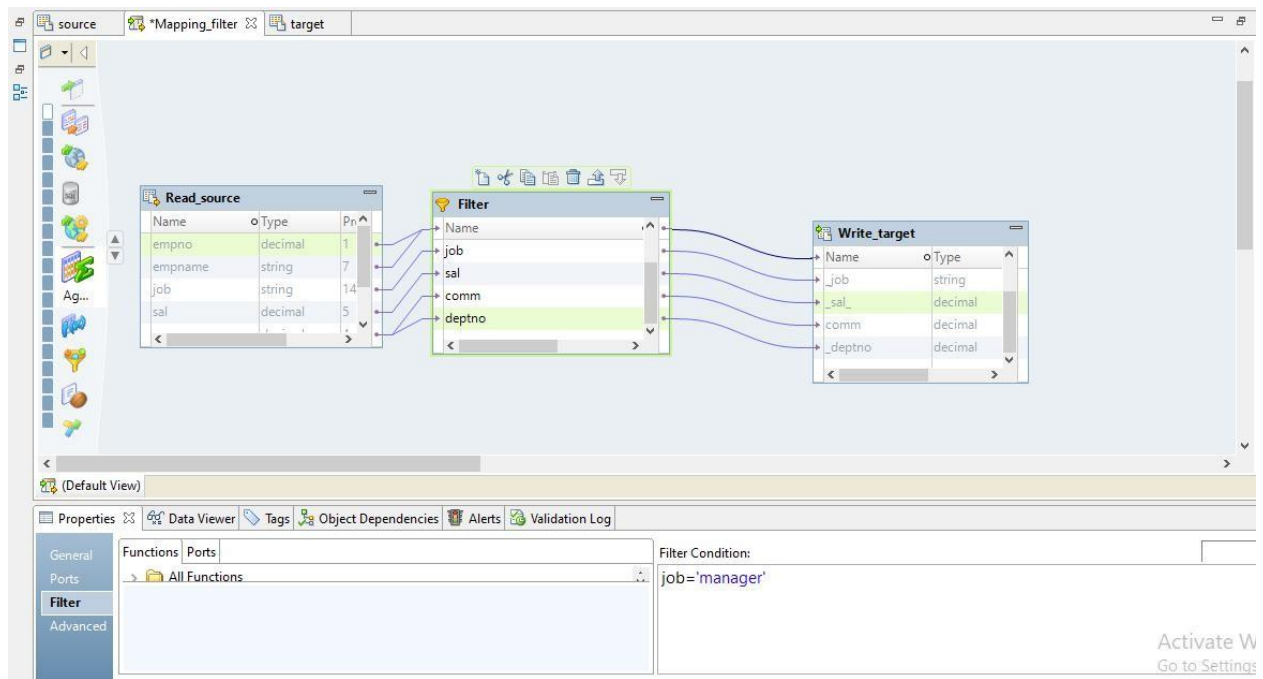
(INPUT):



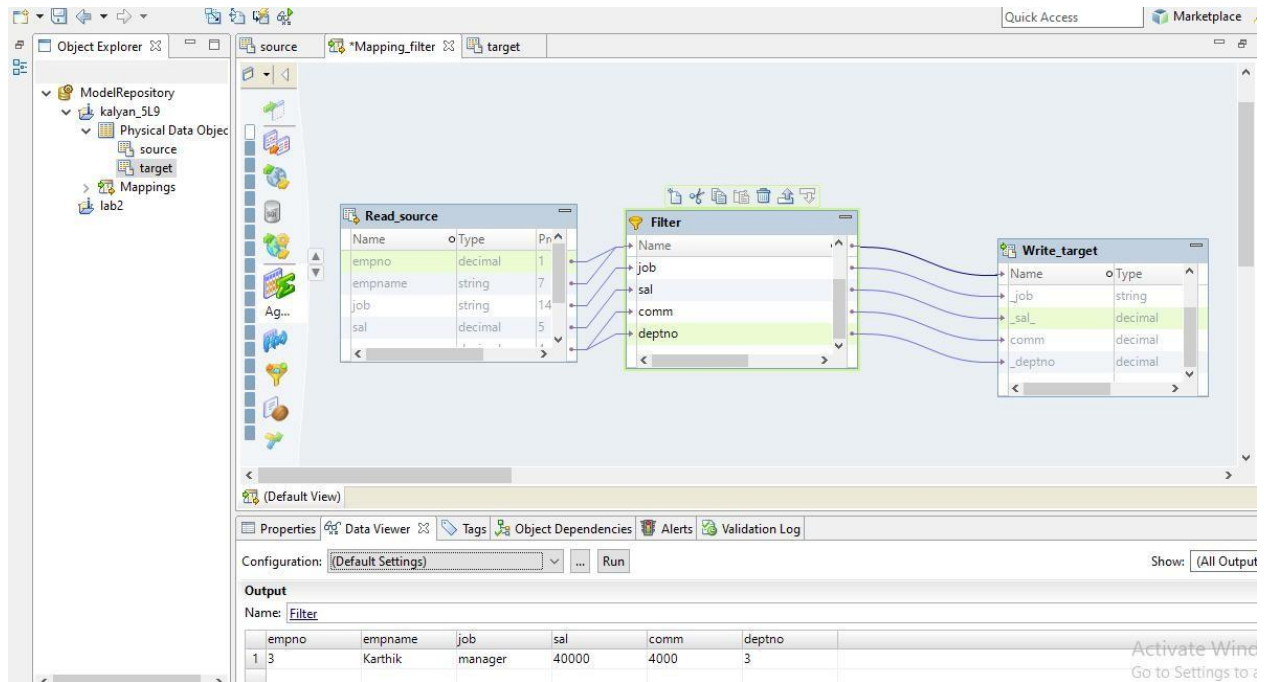
OUTPUT:



3) job='manager'(INPUT):



OUTPUT:



2.Add the commission of 1000 Rs in the Salary field of Employee table using Expression Transformation.

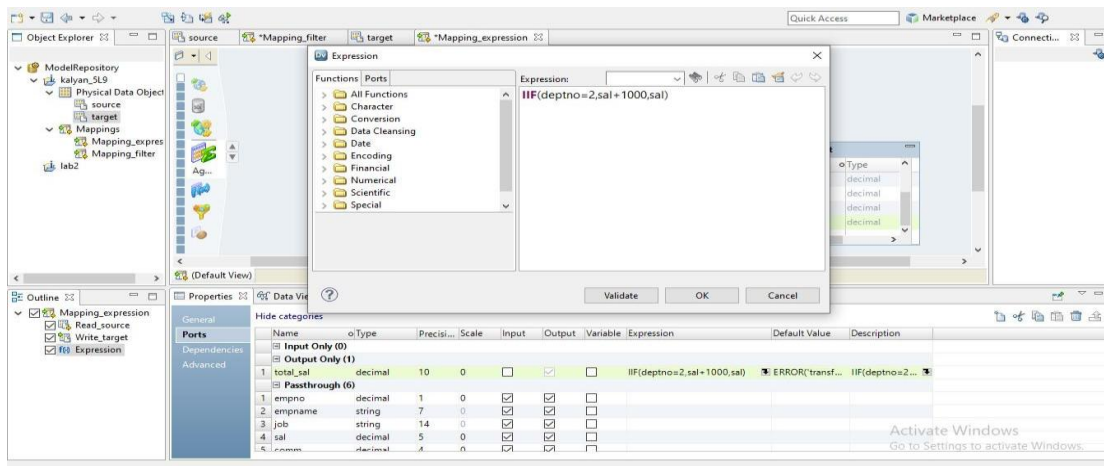
Aim: To apply an Expression Transformation on a flat file to evaluate two columns into an output column in the target table.

Theory: Expression transformation is a connected, passive transformation used to calculate values on a single row. Examples of calculations are concatenating the first and last name, adjusting the employee salaries, converting strings to date etc. Expression transformation can also be used to test conditional statements before passing the data to other transformations.

Procedure:

Step 1: Create the source and target flat files (text files), with the first row being column name and subsequent rows being records, separated by ',' as a delimiter. EXPRESSION:

IIF(deptno=2,sal+1000,sal):



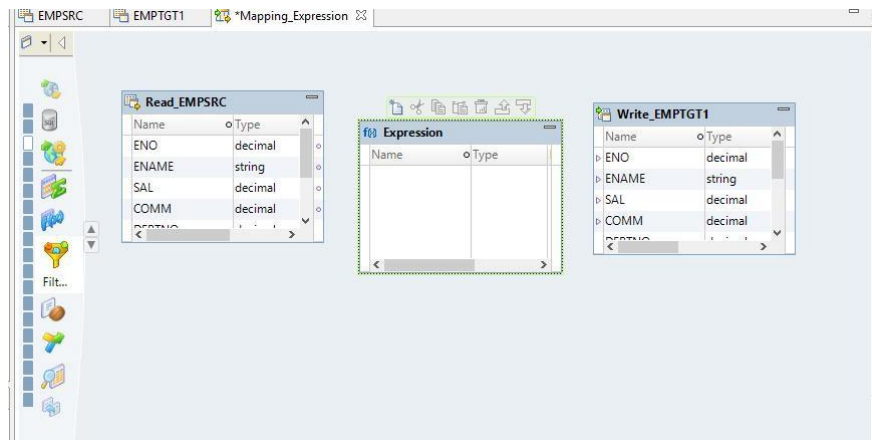
Step 2: Start up Informatica Services and Launch Informatica Developer.

Step 3: Import your flat files (both source and target files) into **Vaishnavi_N1** project.

Click on Next and then Next again. In the next dialog box that opens, check the **"Import Column Names from the first line"** checkbox and make sure the delimiter selected is ','. Then click on **Finish**.

Step 4: Once the files have been imported, create a new mapping and name it **Expression_Mapping**.

Step 5: Drag the source (EMPSRC) table into the workspace (in **read** mode) and the target (EMPTGT1) table (in **write** mode). Also drag the Expression Transformation from the palette.



Step 6: Drag and drop the values of the target table into the **Expression Transformation**. Also link up the input table and the output table via the Expression Transformation appropriately. Make sure that the Transformation is selected. Then navigate to **Properties** → **Ports** in the **Properties Bar**. Unselect the **Input Mark** for **TOTALSAL**. This ensures that it can only act as an output port after evaluation of the expression specified.

Step 7: In the Expression Section of the entry, click on the small arrow head to open up the **Expression Dialog Box** and enter the following expression: “IIF(deptno=2,sal+1000,sal):”

This expression checks if the Commission field is NULL, then returns only salary as the result, else returns Salary + Commission as result to the TOTALSAL field. Don't forget to validate the expression for errors.OUTPUT:

empno	empname	job	sal	comm	deptno	total_sal
1	kalyan	sde	30000	1000	2	31000
2	varun	tester	25000	2000	1	25000
3	Karthik	manager	40000	4000	3	40000
4	Rishi	projectManag...	45000	2100	3	45000
5	sai	poen	21000	1000	4	21000
6	Rithvik	salesMANager	32000	1200	3	32000

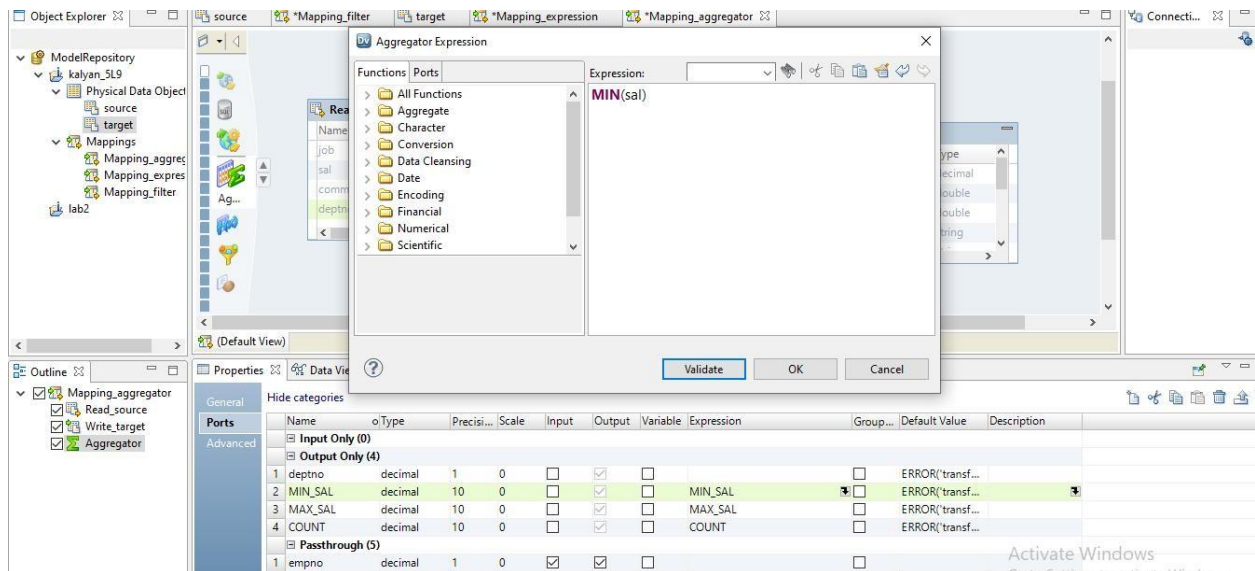
3. Use Aggregator transformation to display the average salary of employees in each department.

Aim: To apply an Aggregator Transformation on a Flat file object to calculate Minimum and Maximum Salary in the table.

Theory: Aggregator transformation is an active transformation used to perform calculations such as sums, averages, counts on groups of data. The integration service stores the data group and row data in aggregate cache. The Aggregator Transformation provides more advantages than the SQL, you can use conditional clauses to filter rows.

Procedure:

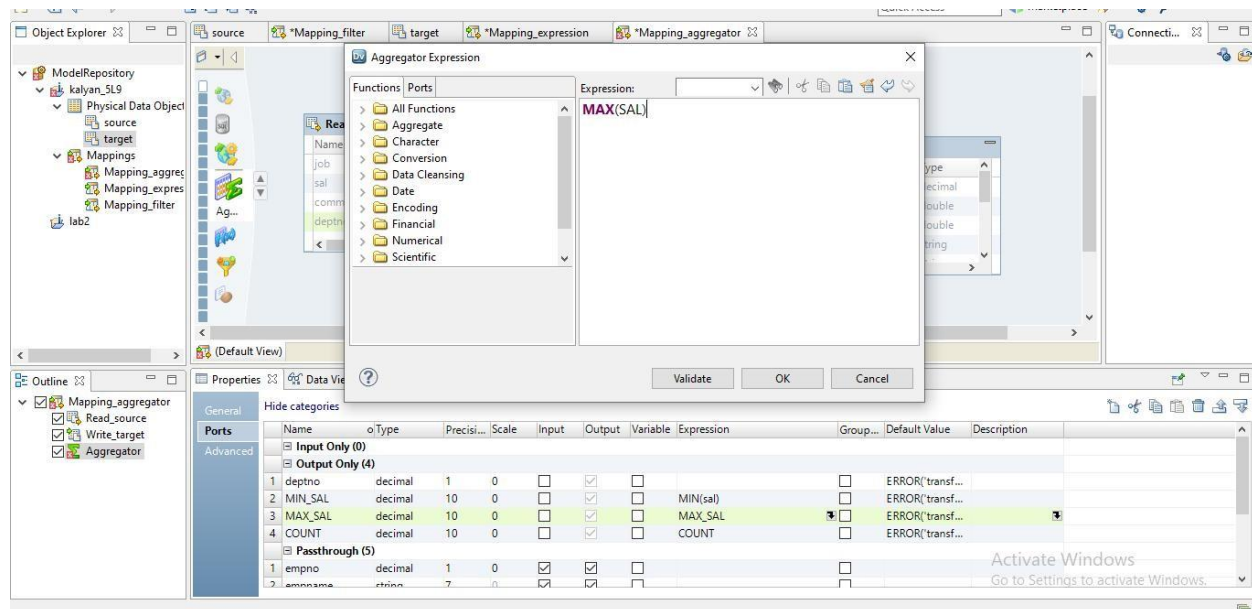
Step 1: Create the source and target flat files (text files), with the first row being column name and subsequent rows being records, separated by ‘,’ as a delimiter. 1)(MIN_SALARY):



Step 2: Start up Informatica Services and Launch Informatica Developer.

Step 3: Import your flat files (both source and target files) into **Vaishnavi_N1** project.

2) (MAX SAL):



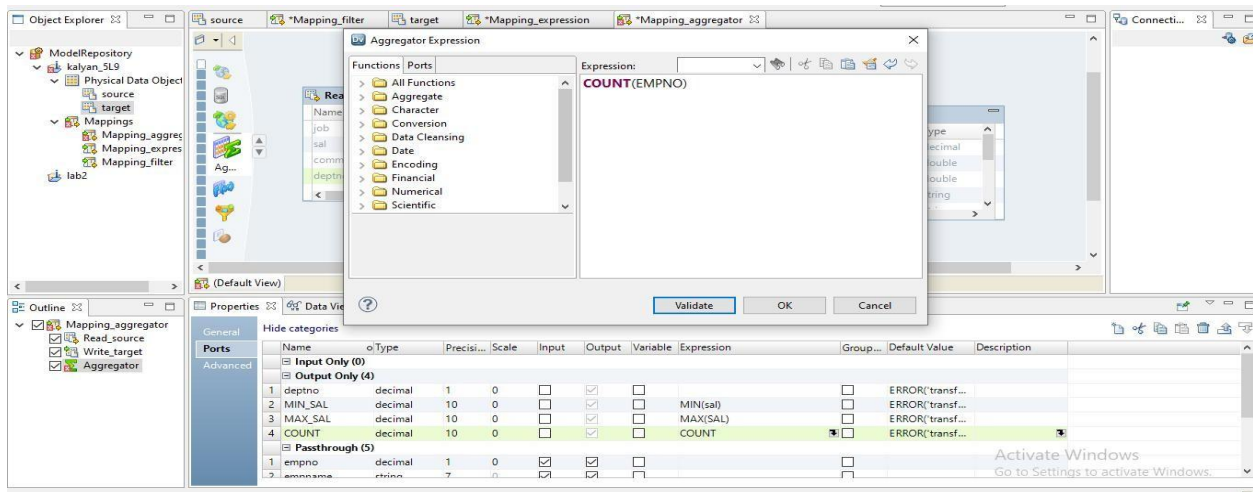
Click on Next and then Next again. In the next dialog box that opens, check the **“Import Column Names from the first line”** checkbox and make sure the delimiter selected is ‘,’. Then click on **Finish**.

Step 4: Once the files have been imported, create a new mapping and name it **Mapping_Aggregator**.

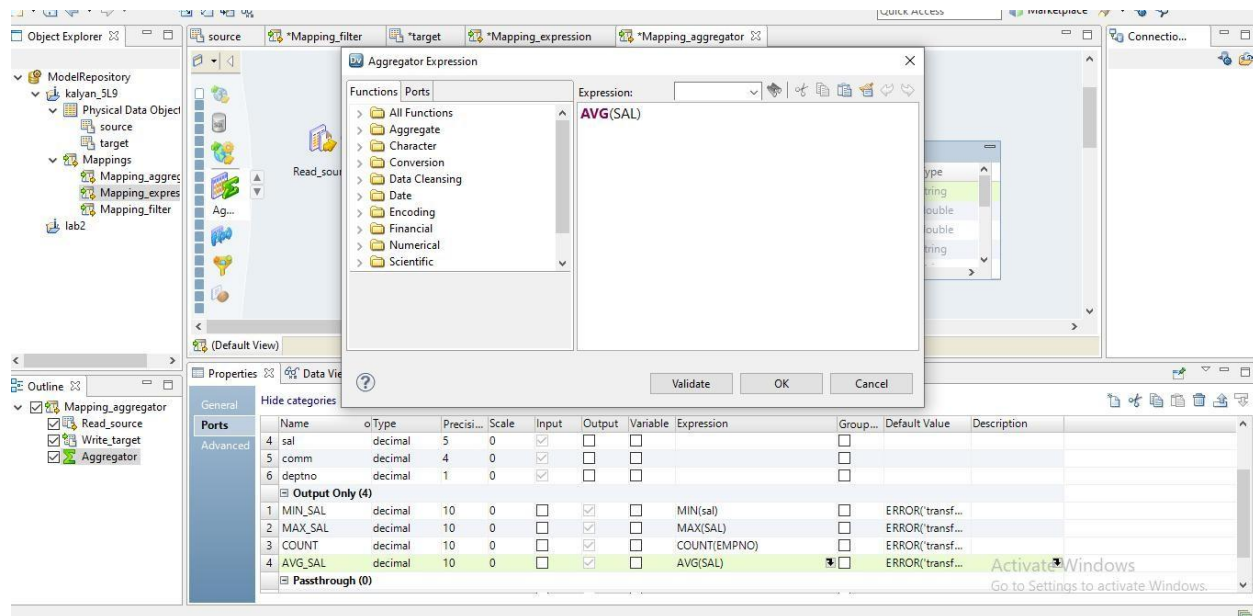
Step 5: Drag the source (EMPSRC) table into the workspace (in **read** mode) and the target (AGGREGATOR_TARGET) table (in **write** mode). Delete all the non-required entries from the tables. Also drag the Aggregator Transformation from the palette. Drag and drop the values of the target table into the **Aggregator Transformation**. Link up the input table and the output table via the Aggregator Transformation appropriately. Add MIN_SAL, MAX_SAL and COUNT entries to the Transformation as they are required for the evaluation.

Step 6: Make sure that the Transformation is selected. Then navigate to **Properties** → **Ports** in the **Properties Bar**. Unselect the **Input Mark** for **MIN_SAL**, **MAX_SAL**. This ensures that it can only act as an output port after evaluation of the expression specified

3) (COUNT):



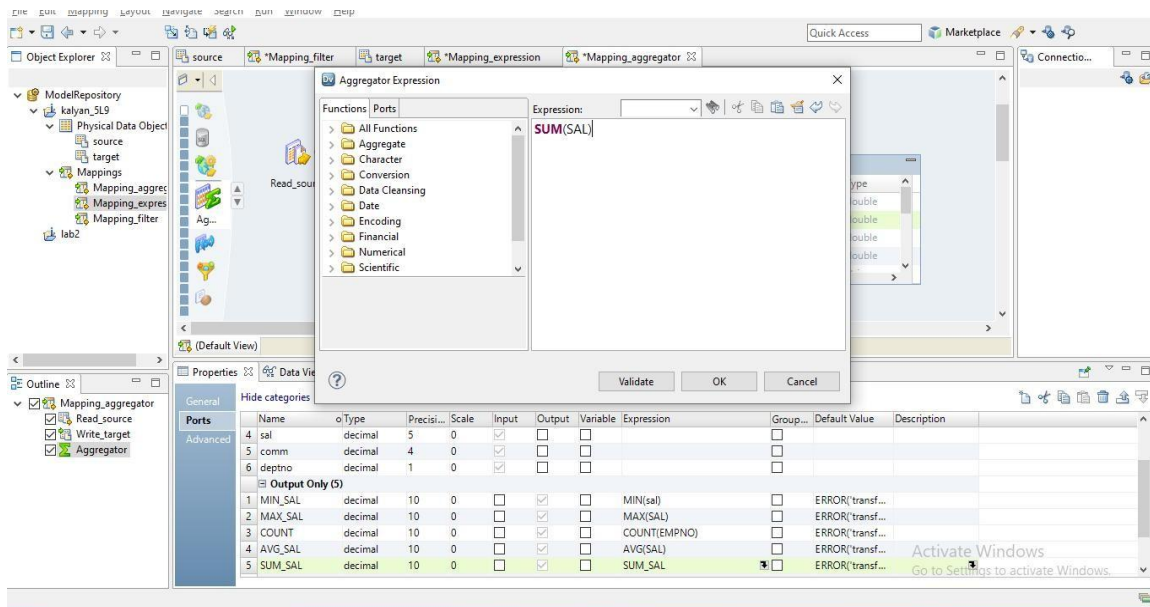
Step 7: In the Expression Section of the MIN_SAL entry, click on the small arrow head to open up the **Expression Dialog Box**, navigate to **Functions** → **Aggregate** and select **MIN** function. This function checks for the least value among all the values and returns it. Then navigate to **Ports** and select **SAL**, so that the expression reads “**MIN(SAL)**”. Don’t forget to validate the expression for errors. 4) (AVERAGE):



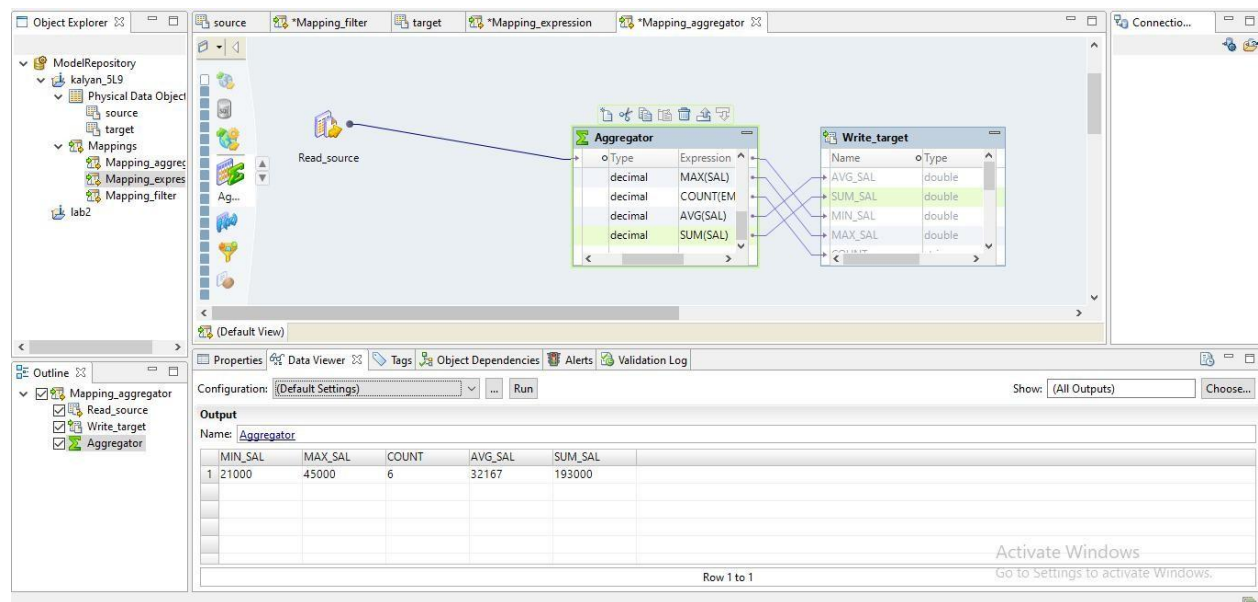
Step 8: Now Run the Mapping by navigating to **Run** → **Run Mapping** in the Menu Bar.

Step 9: Then navigate to “**F:\Informatica\PCExpress\tomcat\bin\target**” (that’s where Informatica is installed on my system, F: Drive), and you can find your output target file. Open it up to see the output.

5) (TOTAL_SAL):



OUTPUT:



4. Use Joiner transformation to display the Sailor_ID (OR) Sailor_Name form Sailors table and Boat_ID (OR) Boat_Name from Boats table in a new table.

Aim: To apply Joiner Transformation on a Data Set to perform Normal Join.

Theory: The joiner transformation provides you the option to create joins in Informatica. The joins created using joiner transformation are similar to the joins in databases. The advantage of joiner transformation is that joins can be created for heterogeneous systems (different databases).

In joiner transformation, there are two sources which we are going to use it for joins. These two sources are called

- . Master Source
- . Detail Source

In the properties of joiner transformation, you can select which data source can be Master and which source can be detail source. During execution, the master source is cached into the memory for joining purpose. So it is recommended to select the source with less number of records as the master source.

The following joins can be created using joiner transformation

1. **Master outer join:** In Master outer join, all records from the Detail source are returned by the join and only matching rows from the master source are returned.
2. **Detail outer join:** In detail outer join only matching rows are returned from the detail source, and all rows from the master source are returned.
3. **Full outer join:** In full outer join, all records from both the sources are returned. Master outer and Detail outer joins are equivalent to left outer joins in SQL.
4. **Normal join:** In normal join only matching rows are returned from both the sources.

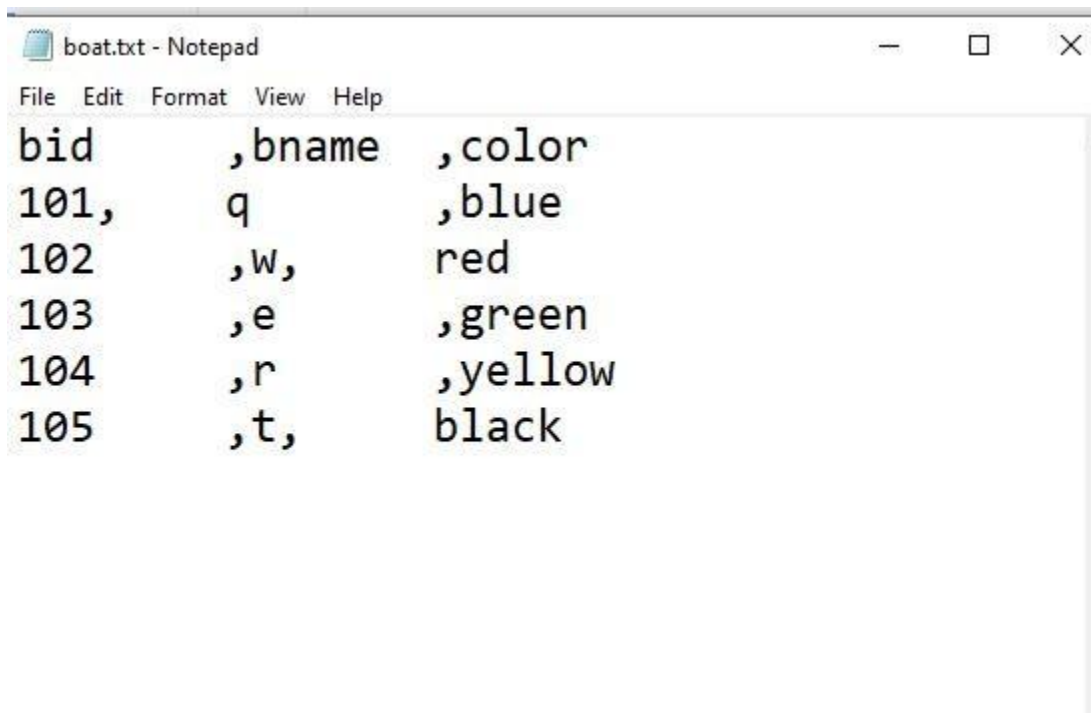
Procedure:

Step 1: Create the following tables Boats(bid number(2), bname varchar2(20), color varchar2(20)), Reserves(sid number(2), bid number(2), day date)), Sailors(sid number(2),sname varchar2(20), age number(2), rating number(2)) and populate them with values.

Step 2: Start up Informatica Services and Launch Informatica Developer.

Step 3: Create a new project (Kalyan_5L9) and import your tables from the database (both source and target files).

BOATS TABLE:



```
boat.txt - Notepad
File Edit Format View Help
bid      ,bname  ,color
101,     q      ,blue
102      ,w,     red
103      ,e      ,green
104      ,r      ,yellow
105      ,t,     black
```

RESERVES TABLE:

```
reserves.txt - Notepad
File Edit Format View Help
sid      ,bid,    date
1        ,101,    30/10/2023
2        ,102,    23/11/2023
3        ,103,    12/09/2023
4        ,104,    18/12/2023
5        ,105,    02/03/2023
```

SAILORS TABLE:

sailors.txt - Notepad

sid,	sname	,age,	rating
1,	a	,30,	7
2,	b	,45,	8
3,	c	,35,	9
4,	d,	40,	6
5,	e,	25,	4

INPUT:

Informatica Developer

File Edit Mapping Layout Navigate Search Run Window Help

Object Explorer

- ModelRepository
 - acd
 - kalyan_5L9
 - Physical Data Ob
 - boat
 - reserves
 - sailors
 - source
 - target
 - Mappings
 - Mapping_agc
 - Mapping_exp
 - Mapping_filt
 - Mapping_joir
 - sruthi_5y6
 - ModelRepository
 - ModelRepository

Outline

- Mapping_Joiner
 - Read_reserves
 - Read_sailors
 - Joiner

Properties

Name: Joiner

Description:

Activate Windows
Go to Settings to activate Windows.

29°C Haze 07:56 21/11/2023

Step 4: Once the files have been imported, create a new mapping and name it Mapping_Joiner.

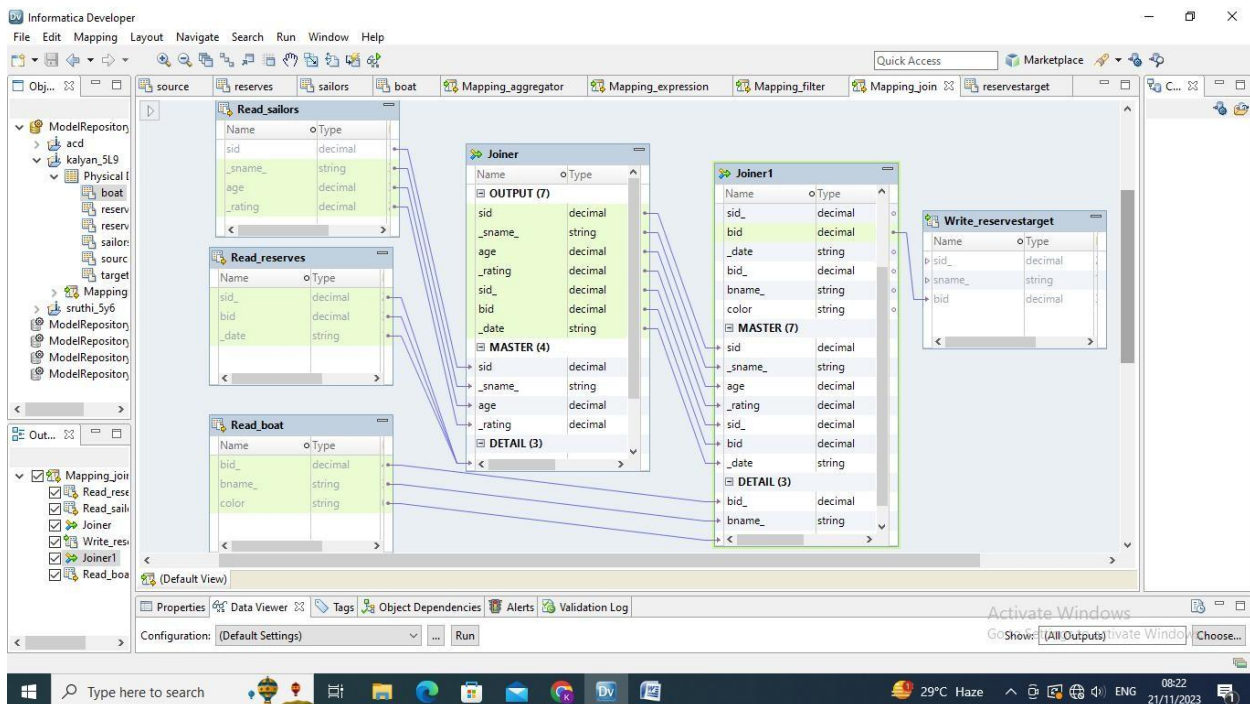
Step 5: Drag all the three tables in read mode and place them in a vertical column.

Step 6: Now drag joiner from the palette and load two tables into it. (Both tables must have a common row)

Step 7: Navigate to Properties Join in the Properties Tab. Specify the join type. In this case, we use Normal Join. And specify the join condition sid of Reserves = sid of Sailors i.e., Master SID = Detail SID1.

Navigate to Properties -> Advanced of the Properties Tab and put a check mark to the Sorted Input option so that the input is sorted.

Step 8: Now Drag another Joiner from the palette and give the output of the previous Joiner as the Master and Boats as the Detail. Give the condition as Master BID = Detail BID1 and Sorted Input as previously done.



Step 9: Now Run the Mapping. Check the Database or the DataViewer for the Output.

OUTPUT:

reservestarget.csv - Microsoft Excel

A1		1															
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	1	101															
2	2	102															
3	3	103															
4	4	104															
5	5	105															
6																	
7																	
8																	
9																	
10																	
11																	

5. FIND RANK OF TOP TWO EMPLOYEES BASED ON THEIR ID'S WITHOUT USING RANK FUNCTION.

Aim: To apply Rank Transformation on a Flat File Data Object to display the top Employee's ID without using RANK function.

Theory: Rank transformation is an active and connected transformation. The rank transformation is used to select the top or bottom rank of data. The rank transformation is used to select the smallest or largest numeric/string values. The integration service caches the input data and then performs the rank calculations.

Procedure:

Step 1: Create the source and target flat files (text files), with the first row being column name and subsequent rows being records, separated by ',' as a delimiter.

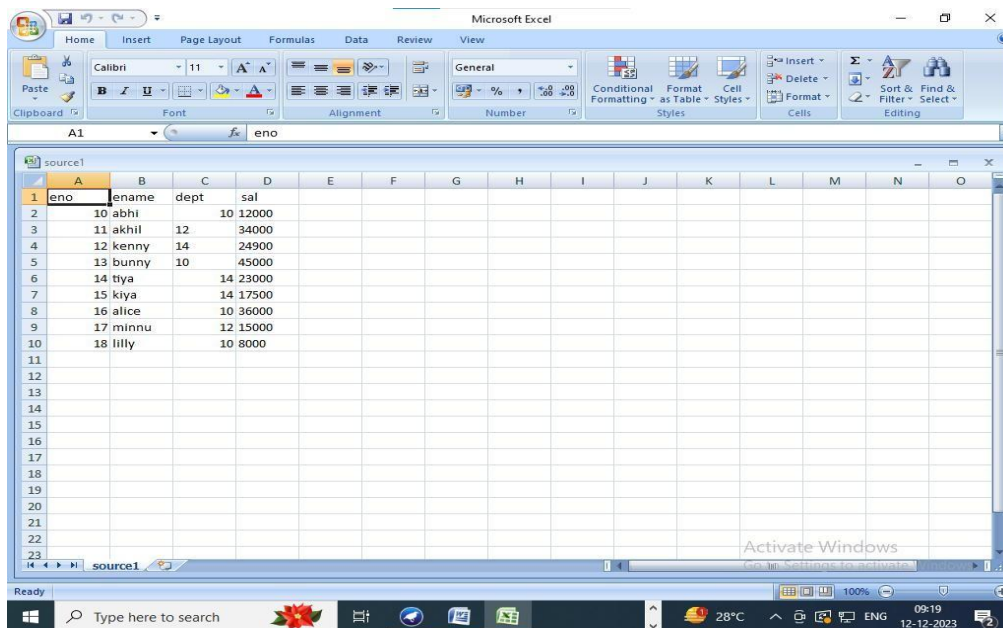
Step 2: Start up Informatica Services and Launch Informatica Developer.

Step 3: Import your flat files (both source and target files) into mirmo5 project. Click on Next and then Next again. In the next dialog box that opens, check the "Import Column Names from the first line" checkbox and make sure the delimiter selected is ','. Then click on Finish.

Step 4: Once the files have been imported, create a new mapping and name it Mapping_Rank

Step 5: Import the source file with read mode and target file as write mode and drag and drop the Rank Transformation from the Palette into the workspace. Give the necessary connection

SOURCE FILES:



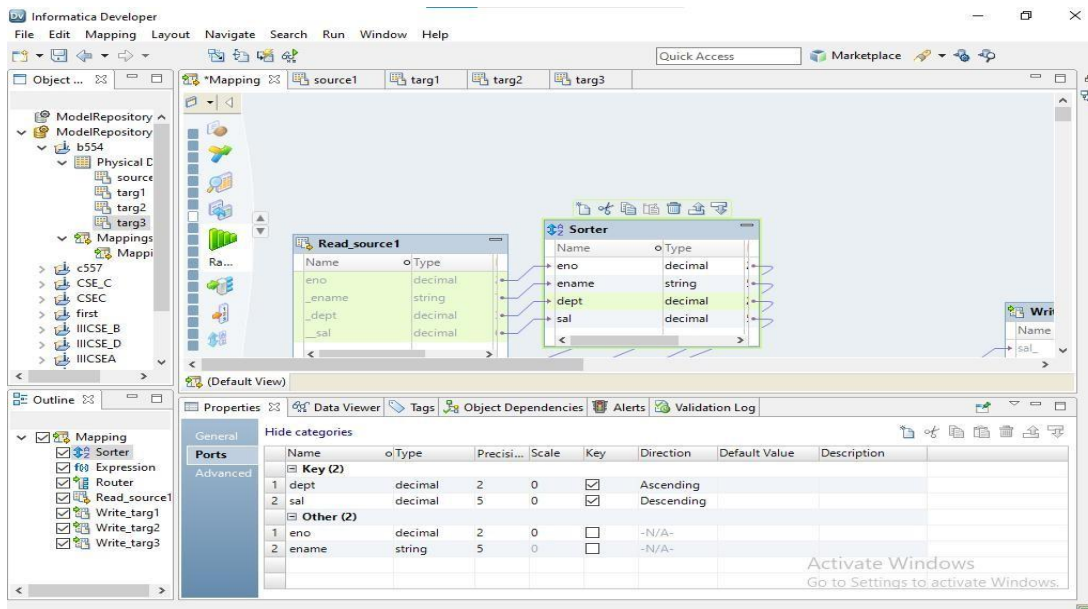
The screenshot shows a Microsoft Excel spreadsheet with the following data:

eno	ename	dept	sal
10	abhi	10	12000
11	akhil	12	34000
12	kenny	14	24900
13	bunny	10	45000
14	ttya	14	23000
15	kiya	14	17500
16	alice	10	36000
17	minnu	12	15000
18	lilly	10	8000

Step 6: Select the SORTER, EXPRESSION, ROUTER Transformation box consecutively.

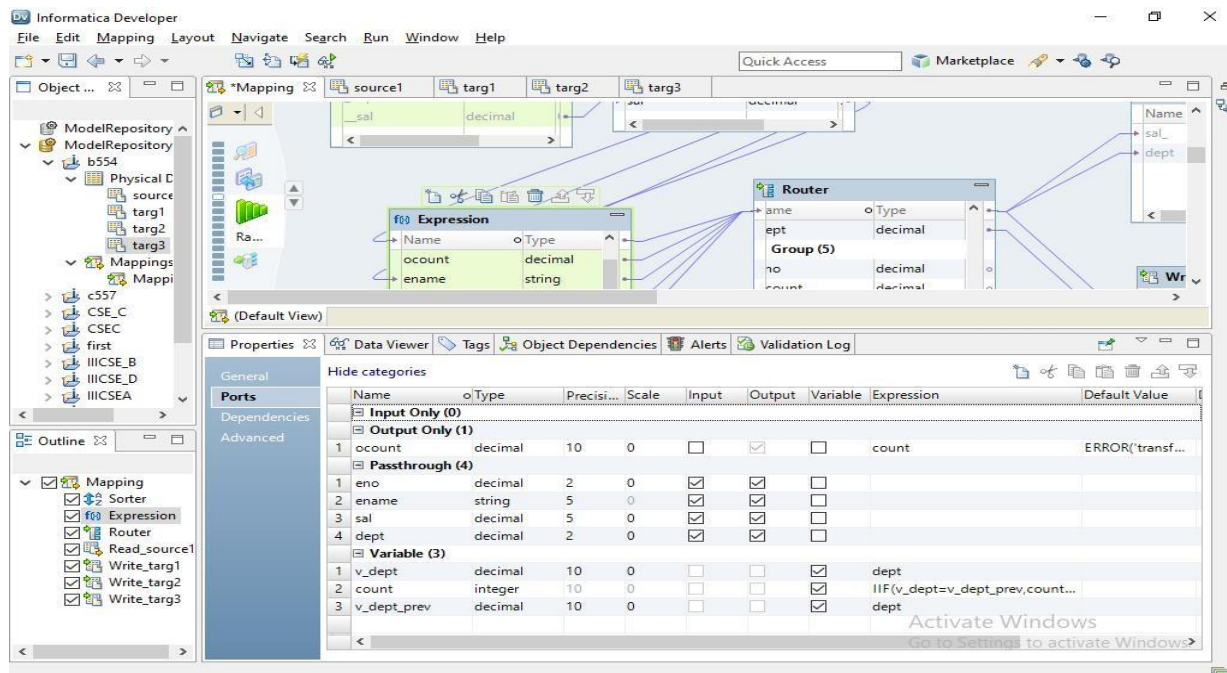
SORTER TRANSFORMATION:

Navigate to port section -> there in that section apply Ascending for dept and Descending order for sal columns remaining two columns are unchecked for the Key condition mentioned in that.



EXPRESSION TRANSFORMATION:

Then after the output of sorter is given as input for the Expression transformation where we need to navigate through ports section there we need to create an extra column named ocount and create three variables v_dept, count, v_dept_prev for each and every variable we need to write an Expression they are for v_dept it is dept and for count it is "IIF(v_dept=v_dept_prev,count+1,1) and for v_dept_prev it is dept.



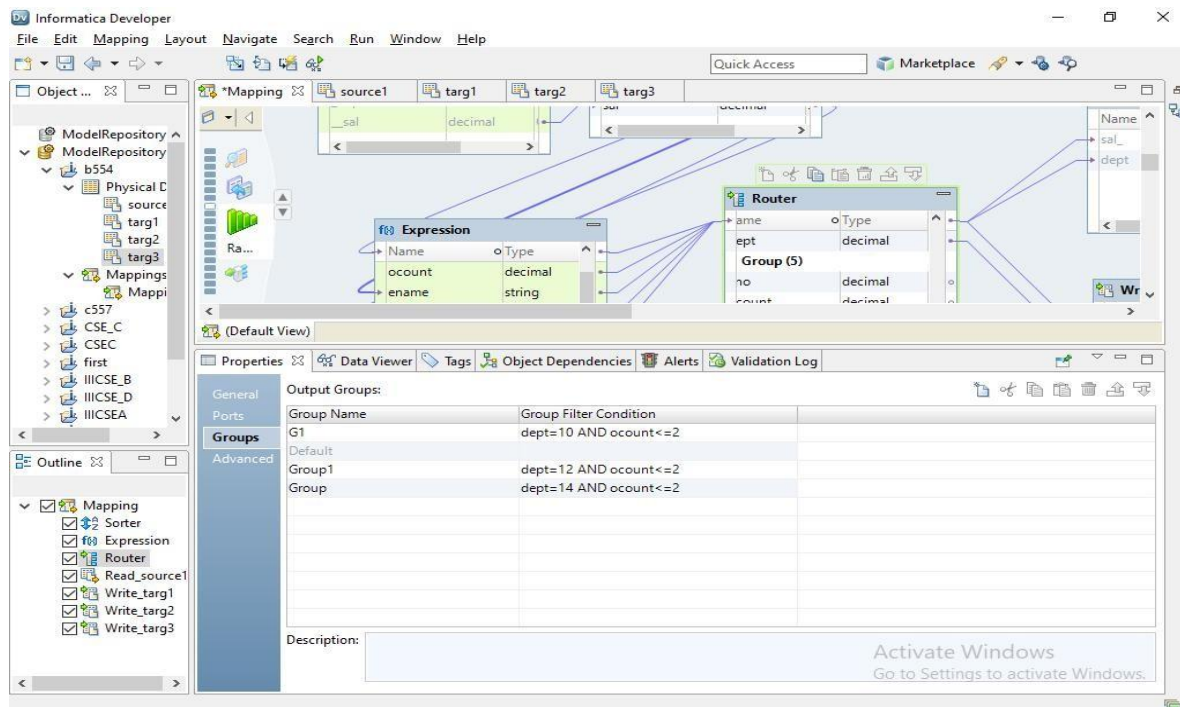
ROUTER TRANSFORMATION:

Then the output of Expression is given as input for router transformation then in this we are going to write some sort of conditions in Groups they are for

G1 – dept=10 AND ocount<=2

Group1 – dept=12 AND ocount<=2

Group – dept=14 AND ocount<=2



Step 8: Make sure that the connections are intact. Now Run the Mapping. Then navigate to "C:\Informatica\PCExpress\tomcat\bin\target" (that's where Informatica is installed on my system, C: Drive), and you can find your output target file. Open it up to see the output. **OUTPUT:**

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	1	45000	10												
2	2	36000	10												
3															
4															
5															
6															
7															
8															
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